

Anything over Dense Wavelength Division Multiplexing with Programmable Data Planes and Software Defined Networking

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Abstract This work proposes using Programmable Data Plane (PDP) switches with Digital Coherent Optical (DCO) pluggables to enable Anything over Dense Wavelength Division Multiplexing (XoDWDM) Open Line Systems (OLSs) with Software-Defined Networking (SDN) control plane. Feasibility is verified in a testbed using hardware PDP switches.

Introduction

Open Line Systems (OLSs) are a key component in modern optical networks, enabling open disaggregated optical networks by eliminating the dependency on proprietary solutions, standardizing the requirements for transmission components, and integrating novel network management paradigms based on Software-Defined Networking (SDN) and open standards. These changes make communication services in OLSs a multi-vendor, interoperable ecosystem with cost-efficient equipment, flexible configuration, adaptability to future requirements, and openness to innovation.

Fig. (1a) shows an exemplary OLS interconnecting two datacenters using Reconfigurable Optical Add/Drop Multiplexers (ROADMs). Edge ROADMs have a small form-factor chassis and connect to datacenters using gray-optical pluggables. Core ROADMs have a larger form-factor chassis, connect to the edge ROADMs using optical fibers spanning hundreds of kilometers, and do not perform Optical-Electrical-Optical (OEO) conversion, therefore forming a transparent OLS.

In transparent OLSs, packets are forwarded based on the optical transport information configured in the ROADMs. Optical forwarding is completely decoupled from packet header information, thereby separating the physical (Layer 1) and Internet (Layer 3) layers. In OLS based on IP over Dense Wavelength Division Multiplexing (IPoDWDM) of Fig. (1b), the physical and Internet layers are coupled. In IPoDWDM, edge and core ROADMs are replaced with edge and core IP routers with ZR/ZR+ Digital Coherent Optics (DCO) pluggables and multiplexers. Edge routers have gray-optical pluggables for connecting to datacenters. Routers perform OEO conversion, resulting in an opaque OLS, in which signals are regenerated at every node and forwarding is based on information from the physical and Internet layers.

This work proposes the OLS with SDN control plane of Fig. (1c) for Anything over Dense Wavelength Division Multiplexing (XoDWDM). This ap-

proach replaces routers in IPoDWDM with Programmable Data Plane (PDP) switches, thereby bringing PDP's network programmability to OLSs. In XoDWDM, forwarding decisions consider information in standard or custom headers from the physical, data link, network, and transport layers (Layers 1-4). Moreover, PDPs are native SDN devices with standardized interfaces for interaction between the control and data planes, thereby enabling seamless integration of SDN into OLSs.

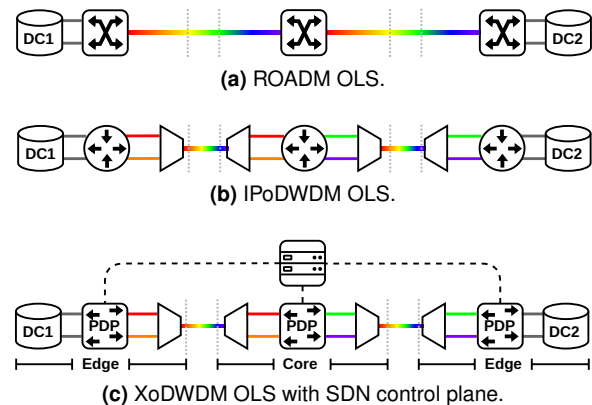


Fig. 1: OLS architectures interconnecting two datacenters.

Pan et al.^[1] presented an IPoDWDM OLS collocating PDP switches and ROADMs. The author used In-band Network Telemetry (INT) to collect telemetry data for flows and encryption to ensure privacy. Dicorato et al.^[2] proposed an SDN architecture for IPoDWDM OLSs using routers and DCO pluggables and evaluated it using a real network testbed. Kamalzadeh et al.^[3] presented an IPoDWDM OLS collocating routers and ROADMs and introduced a hierarchical SDN control architecture based on OpenROADM. Gonzales de Dios et al.^[4] proposed a SDN control architecture for IPoDWDM OLSs with open standard APIs and analyzed the role of artificial intelligence in its automation. To our knowledge, this work is the first to use PDP switches with DCO pluggables to enable XoDWDM OLSs with SDN control plane.

Implementation

The architecture in Fig. (1c) of an SDN OLS for XoDWDM is realized using the Intel Tofino as PDP switch. There are two Tofino versions. The Tofino 1 is an Edgecore DCS801^[5] switch, with a switching capacity of 6.4 Tbps and 32 Quad Small Form-factor Pluggable (QSFP28) ports, supporting ZR/ZR+ DCO pluggables with throughput up to 100 Gbps and reach up to 80 km. The Tofino 2 is an Edgecore DCS810^[6] switch, with a switching capacity of 12.8 Tbps and 32 QSFP-DD ports, supporting ZR/ZR+ DCO pluggables with throughput up to 400 Gbps and reach up to 400 km. Tofino 1 and 2 are programmed in Programming Protocol-independent Packet Processors (P4) language.

The *anything* capacity in XoDWDM is demonstrated by running a P4 program on the Tofino switch that forwards traffic using information from the physical, data link, network, and transport layers, i.e., Layers 1, 2, 3 and 4, respectively. The following fields can be matched in the P4 program: in Layer 1, the ingress port; in Layer 2, the source and destination MAC addresses; in Layer 3, the source and destination IP addresses; and in Layer 4, the source and destination TCP ports. The Tofino switch is configured by the SDN controller using rules that match any of the fields mentioned above and set the output port. Hence, setting the wavelength for outgoing traffic (i.e., output port), based on the input wavelength (i.e., incoming port) and Layer 2-4 information. Although the P4 program uses standard fields in Layers 1-4, it is also possible to include custom fields to store application-specific information, e.g., wavelength, Optical Signal-to-Noise Ratio (OSNR), etc.

Evaluation

Fig. (2) shows the conceptual evaluation testbed focusing on the core node of the OLS architectures in Fig. (1). For each architecture, it is assumed that the same devices are used for edge and core nodes, with the only difference that edge nodes include gray-optical pluggables to connect data-center equipment. Hence, across architectures, the results generalize to core and edge nodes.

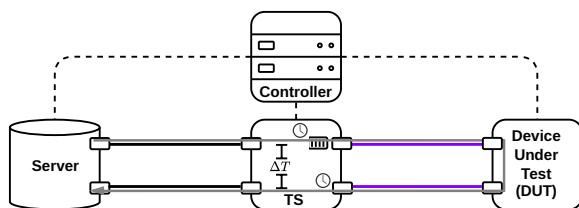


Fig. 2: Conceptual evaluation testbed.

Fig. (3) shows the devices and connections of the testbed. During the evaluation, the *Server* replays a Packet Capture (PCap) file containing 1 million unique 84-byte packets using *tcpreplay* at 10 K packets per second (pps), following the gray arrow in Fig. (2), using 25 Gbps gray-optical

pluggables (pink lines). The packets arrive at the Tofino 1 Time Stamper switch (*TS* in yellow), which stores the time when the packet enters the egress pipeline in a custom header. The *Device Under Tests* (*DUTs*) can be: a Layer 1 Polatis Series 6000 Optical Switch^[7] (*OS* in blue), an Edgecore DCS204 Layer 2-3 switch^[8] (*Router* in orange), an Intel Tofino 1 switch (*Tofino* in green) or a 12 m loopback fiber (in yellow). The *DUT* connects to the *TS* with 100 Gbps 4.5 W QSFP28 pluggables for 10 km at 1310 nm and 6 m single-mode duplex LC/UPC fibers. Baselines for ROADM and IPoDWDM OLS are created with the *OS* and the *Router* forwarding at Layers 1 and 3, respectively. For an XoDWDM OLS, measurements are taken with the *Tofino*, forwarding at the physical, data link, network, and transport layers (i.e., Layers 1-4). The results for the loopback fiber are used to remove the *TS* queuing and forwarding delays in the egress pipeline. Once a packet traverses the *DUT*, it reaches the *TS*, which stores in the custom header the time when the packet reached the ingress interface and the time delta ΔT between the two timestamps, i.e., the evaluation metric representing the packet forwarding time by the *DUT*. The packets are sent to the *Server*, where they are stored in an PCap file for later processing. The *Controller* runs a Python application that deploys the evaluation configurations to the *Server*, *TS* and *DUTs* over an out-of-band control channel (in white). Only one *DUT* is evaluated at a time, and no background traffic is sent to benchmark the devices' best possible performance.

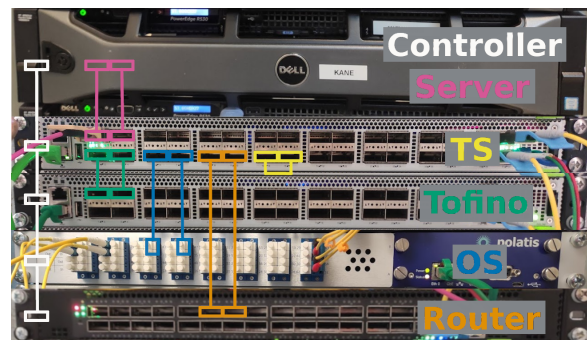


Fig. 3: Evaluation testbed devices and connections.

Fig. (4) and Tbl. 1 show the packet forwarding time ΔT of the *DUTs* to forward a packet for Layers 1-4. The minimum packet forwarding time obtained with the 12 m loopback fiber was subtracted from the results to remove the queuing and forwarding delays of *TS*. The *OS* is the fastest device, with an average packet forwarding time of 48 ns, due to an internal fiber of about 8 m between the input and output ports. The *Router* is the slowest device, with an average packet-forwarding time of 969 ns, due to OEO conversions and Layer 3 routing. The *Tofino* has an average packet forwarding time of 493 ns in Layer 1, which is about 10 times more

than the *OS*. The *Tofino* has an average packet forwarding time of 513 ns in Layer 3, which is about half the time of the *Router*. The packet forwarding time in the *Tofino* is caused by the OEO conversions and the forwarding functions in the given layer. As the layer increases from 1 to 4, the packet forwarding time in the *Tofino* increases by about 10 ns per layer, because the higher the layer, the more headers are parsed by the *Tofino*. E.g., for Layer 4, the standard metadata, Ethernet, IP, and TCP headers are parsed by the *Tofino*.

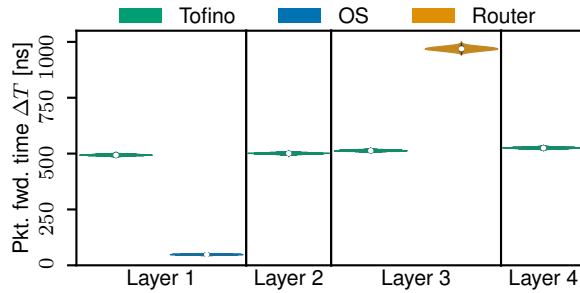


Fig. 4: DUTs packet forwarding time ΔT for Layers 1-4.

Tab. 1: Minimum, average and maximum values for DUTs packet forwarding time ΔT across Layers 1-4.

Device	Layer	Min. [ns]	Avg. [ns]	Max. [ns]
Tofino	1	480	493	508
OS	1	41	48	56
Tofino	2	486	501	514
Tofino	3	501	513	527
Router	3	940	969	1001
Tofino	4	511	525	539

The *OS* is the baseline for a ROADM OLS. It is expected to forward packets the fastest since it is a transparent OLS, where forwarding consists of redirecting the optical signal from the input to the output port. However, in a ROADM OLS, the information in the packet headers cannot be used for forwarding decisions, and there is no signal regeneration at the nodes, since there is no OEO conversion. The *Router* is the baseline for a IPoDWDM OLS. It is expected to forward packets slower than the ROADM OLS, since it is an opaque OLS, resulting in delays due to OEO conversions and Layer 3 forwarding decisions. However, IPoDWDM OLS profits from information in IP or MultiProtocol Label Switching (MPLS) headers for forwarding decisions and signal regeneration at the nodes. The *Tofino* presents the results for the proposed XoDWDM OLS, which enhances IPoDWDM OLS by adding the network programmable capabilities and SDN readiness of PDPs. In a XoDWDM OLS, forwarding decisions use information of any protocol header in Layer 1-4. Moreover, it is possible to define new headers to carry relevant information for any use case. E.g., in INT, a header could keep track of the nodes OSNR, forwarding delay, and

queue size to monitor the OLS and apply corrections. The packet forwarding time in a XoDWDM OLS cannot outperform a ROADM OLS due to the delays incurred by OEO conversion and forwarding decisions. However, a XoDWDM OLS using a *Tofino* can outperform a IPoDWDM OLS using a commercial Layer 2-3 switch performing Layer 3 packet forwarding.

The evaluations used 100 Gbps 4.5 W QSFP28 pluggables for 10 km, since no ZR/ZR+ DCO pluggables were available in the lab. ZR DCO pluggables for 80 km consume up to 5.5 W. According to the *Tofino* 1 switch datasheet, the device has standard QSFP28 interfaces that support up to 6 W and the manufacturer's web page lists compatible 100G QSFP28 ZR4 DCO pluggables^[9]. The presented evaluation for the packet forwarding time of the DUTs generalizes to the ZR/ZR+ DCO pluggables, as the dominating delay, which is caused by the forwarding decisions in the DUTs, is independent of the QSFP28 pluggables. Moreover, the OEO conversion delay across the QSFP28 pluggables is comparable.

Conclusions

This work presents Anything over Dense Wavelength Division Multiplexing (XoDWDM), an Open Line System (OLS) architecture that replaces routers in IP over Dense Wavelength Division Multiplexing (IPoDWDM) with Programmable Data Plane (PDP) switches equipped with Digital Coherent Optics (DCO) pluggables. This architectural shift integrates the network programmable capabilities and Software-Defined Networking (SDN) readiness of PDPs into OLSs. In XoDWDM OLS, forwarding decisions can be based on header fields in Layers 1-4, or custom networking applications, such as In-band Network Telemetry (INT), can be easily introduced. Moreover, SDN enables a flexible deployment and configuration of the OLS network applications. The feasibility of XoDWDM OLS was demonstrated through a hardware testbed using an Intel *Tofino* 1 switch. Experimental results show that the XoDWDM OLS introduces higher packet forwarding latency compared to transparent Reconfigurable Optical Add/Drop Multiplexer (ROADM) OLS, but outperforms conventional opaque IPoDWDM OLS using a commercial switch with Layer 3 forwarding. Packet forwarding delay in XoDWDM OLS was about half of that observed in IPoDWDM OLS, while allowing forwarding decisions in Layers 1-4. The support for the Intel *Tofino* switch for DCO was verified using information provided by the manufacturer on compatible Quad Small Form-factor Pluggable QSFP28 modules. Future work will focus on integrating ZR/ZR+ DCO pluggables into the testbed and leveraging XoDWDM OLSs to enable sustainable and resilient future optical core networks.

Acknowledgements

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